

AZ GROUP

CHEMISTRY PAID MIDNIGHT PAPER

2nd Year | For Pre-Medical & Pre-Engineering Students

SECTION B (SHORT QUESTIONS)

INORGANIC CHEMISTRY

Q2. i) Describe the group trend of atomic radius, Ionization energy and electronegativity in S-Block.

ii) Define Transition elements. Describe any two of the following properties:
Variable oxidation states, Color complexes, Magnetic properties.

iii) What is diagonal relationship? Describe the diagonal relationship between any two of the following:

Li & Mg, B & Si, Be & Al

iv) Definitions:

- Homologous series, Isomerism, Polymerization, Catenation
- Chiral carbon, Anti-pyretic medicine, Analgesics, Bucky ball

v) Write the I.U.P.A.C. names of any FOUR of the following complexes:

- $[\text{Cr}(\text{en})_3](\text{NO}_3)_3$
- $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]^{+2}$
- $[\text{Zn}(\text{OH})_2]^{-2}$
- $\text{Na}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$
- $[\text{Cu}(\text{en})_2]^{+2}$
- $[\text{Ni}(\text{SCN})_4]^{-2}$
- $[\text{Ag}(\text{NH}_3)_2]\text{OH}$
- $\text{K}_3[\text{Fe}(\text{CN})_6]$

vi) What are synthetic polymers? Explain PVC OR Nylon. **OR**
Describe Ozone layer. Also write the causes of its depletion.

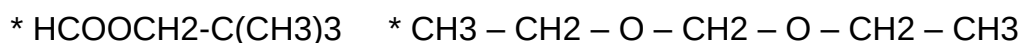
vii) What is spectrometry? Describe one use of each of the following:

ORGANIC CHEMISTRY

viii) Describe homologous series of organic compounds and write three properties of homologous series.

ix) Draw and explain the orbital structure of Ethene **OR** Ethyne and write two methods of its preparation.

x) Write the IUPAC naming of following organic compounds(CAN DIFFER ALSO SEE BOOK IUPAC)



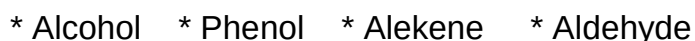
OR Write equations for the following reactions:

- Dehydrohalogenation of Ethyl chloride
- Ozonolysis of Ethene
- Dehalogenation of 1, 1, 2, 2-Tetrabromo ethane
- Hydration of ethyne under specific conditions

xi) What is Friedel-Craft Alkylation and Friedel-Craft Acylation? Also give their mechanism.

xii) Organometallic compounds: Write the reaction of Methyl Magnesium Iodide with CH_3NH_2 , H_2O , and CO_2 .

xiii) Name four derivatives of Carboxylic acid and write the equation for their preparation. **OR** Identify each of the following with one laboratory test



SECTION C (LONG QUESTIONS)

INORGANIC CHEMISTRY

Q3. Explain with the help of a diagram of Castner-Kellner cell, how caustic soda is obtained by the electrolysis of aqueous sodium chloride?

OR

Write the balanced equations for the following chemical processes:

1. A piece of aluminum is dipped into concentrated sulphuric acid.
2. Ferric chloride is mixed with an aqueous solution of caustic soda.
3. Sodium burns in excess of air.

4. Magnesium is heated with nitrogen gas.
5. Potassium is put into ethyl alcohol.
6. Chlorine gas is passed through an aqueous solution of caustic soda.

OR

What is bleaching powder? How is it prepared? Give the reaction of bleaching powder with water and hydrochloric acid.

Q4. What is meant by acid rain? Describe its causes and adverse effects on human lifestyle.

OR

Describe how 99.99% pure copper is obtained from its chalcopryite ore.

Q5. Define troposphere and stratosphere. Also explain the chemistry involved due to the presence of oxides of carbon and nitrogen in the troposphere.

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Q7. What is meant by orientation of benzene? Explain ortho-para and meta directing groups. Write the equations for the preparation of TNT and m-nitro toluene from benzene.

Q8. What is Beta-elimination? Discuss the mechanisms of E1 and E2 reactions. OR

What are Proteins? Classify various types of proteins on the basis of their function.

Q9. Write the structural formulae of any EIGHT of the following:

Hydroquinone, Diphenyl methanol, ter-Butyl Benzoate, Adipic acid, Butanoyl bromide, neo-Pentyl iodide, iso-valeric acid, iso-propyl ter-pentyl ether, Divinyl acetylene, Benzophenone.

OR

Write the equations for any EIGHT of the following chemical processes:

- Addition of HCN in Aldehyde
- Oxidative-Cleavage of Ethylene glycol
- Clemmensen reduction reaction
- Oxidation of Phenol
- Dow's process
- Hydrolysis of Benzene Diazonium salt
- Oxidation of sec. Alcohol with PCC
- Reduction of Carboxylic acid
- Williamson synthesis of ether
- Ozonolysis of 2,3-dimethyl-2-butene

- Dehydration of excess of Alcohol
- Phenol is heated with cone. Nitric acid

End of Paper. BEST of LUCK from **AZ Group**

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